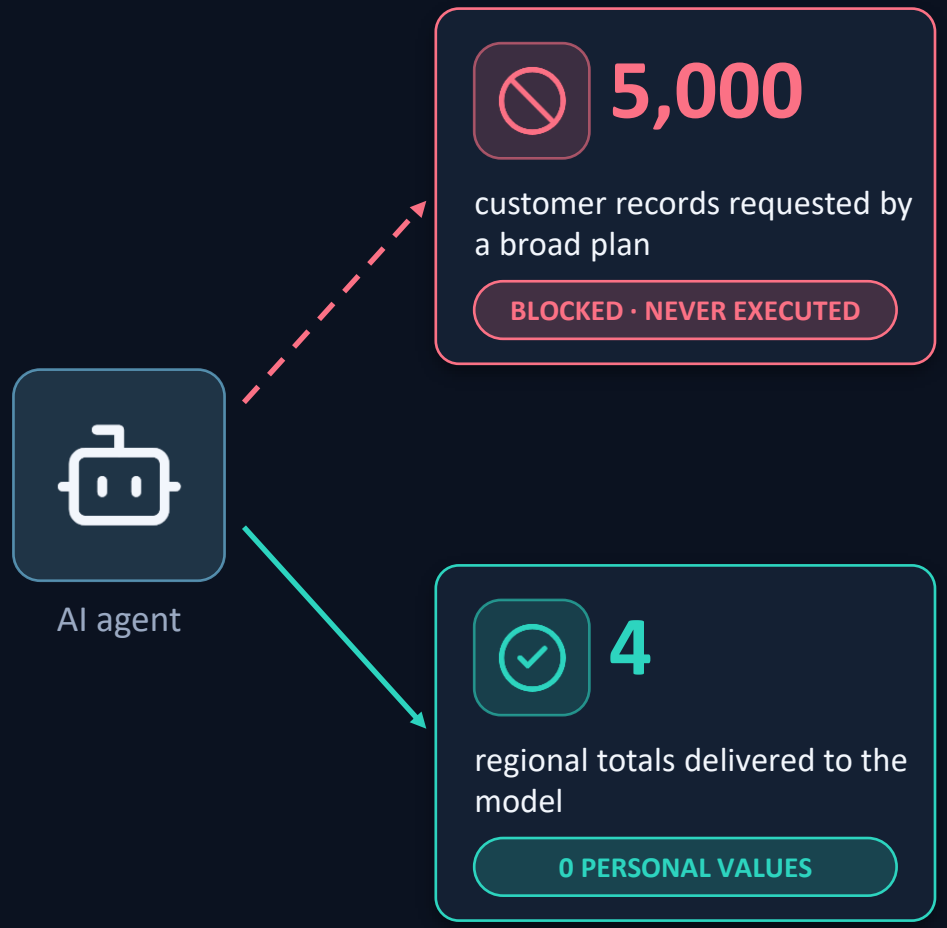
Your AI agent needs an answer. Not your entire database.

ScopeGuard

The intelligent data firewall for MCP-powered agents

Measured on a synthetic 5,000-customer CRM with real MCP tool calls.



One simple question. One very broad tool.

The user asks

“Summarize September sales by region.”

What the task needs

4 regional totals

North · South · East · West

What a broad CRM tool call requests

```
crm.get_customers(
   fields=[region, email, synthetic_personal_id,
           total_spend_cents],
   month="2026-09", limit=5000)
```

`region` `email` `synthetic_personal_id` `total_spend_cents`

5,000
customer rows requested

10,000
personal values requested

max class: PERSONAL

The task needs regional totals; the tool exposes thousands of customer records.

Block, replan, complete

Traditional gateway

Block and stop

Broad request → Blocked → Task fails

ScopeGuard

Block, replan, complete

Broad request → Blocked → Replanned → Answered

1. Hard policy

Admin-owned rules outside the LLM. Secrets are never disclosed.

2. AI candidate planning

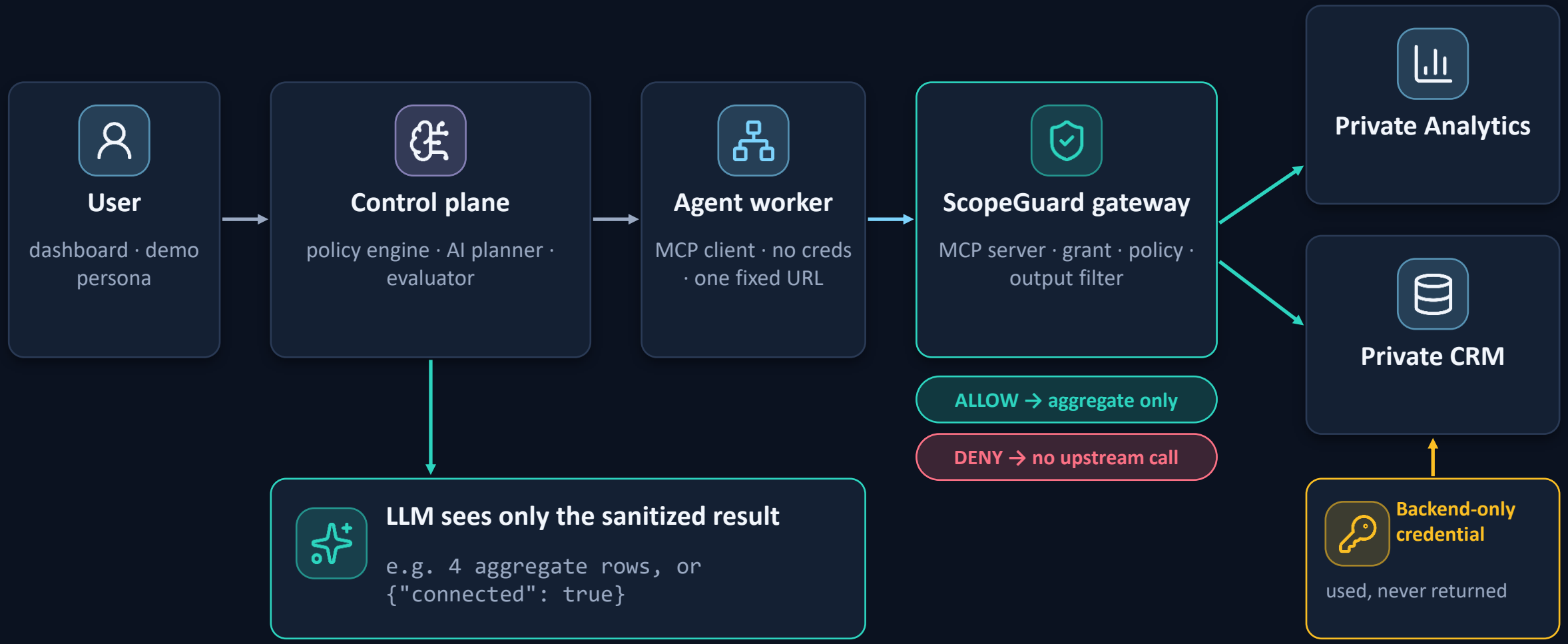
The LLM proposes plans from an approved catalog. It never authorizes.

3. Runtime enforcement

Every MCP call and every result is checked before the model sees it.

Security that helps the agent finish its job safely.

How it works: agents only ever talk to ScopeGuard



Worker → gateway: MCP Streamable HTTP · gateway → private servers: MCP stdio · registered tools only

Three candidate plans, one safe choice

Plan	Tool call	Data requested	Rows (est.)	Personal	Decision
A	crm.get_customers	region, email, personal_id, spend	5,000	10,000	DENY — not executed
B	analytics.get_transactions	region, amount_cents	5,000	0	MORE EXPOSURE — denied for this task
C	analytics.get_sales_summary	region, value_cents, tx_count	4 agg.	0	SELECTED — executed

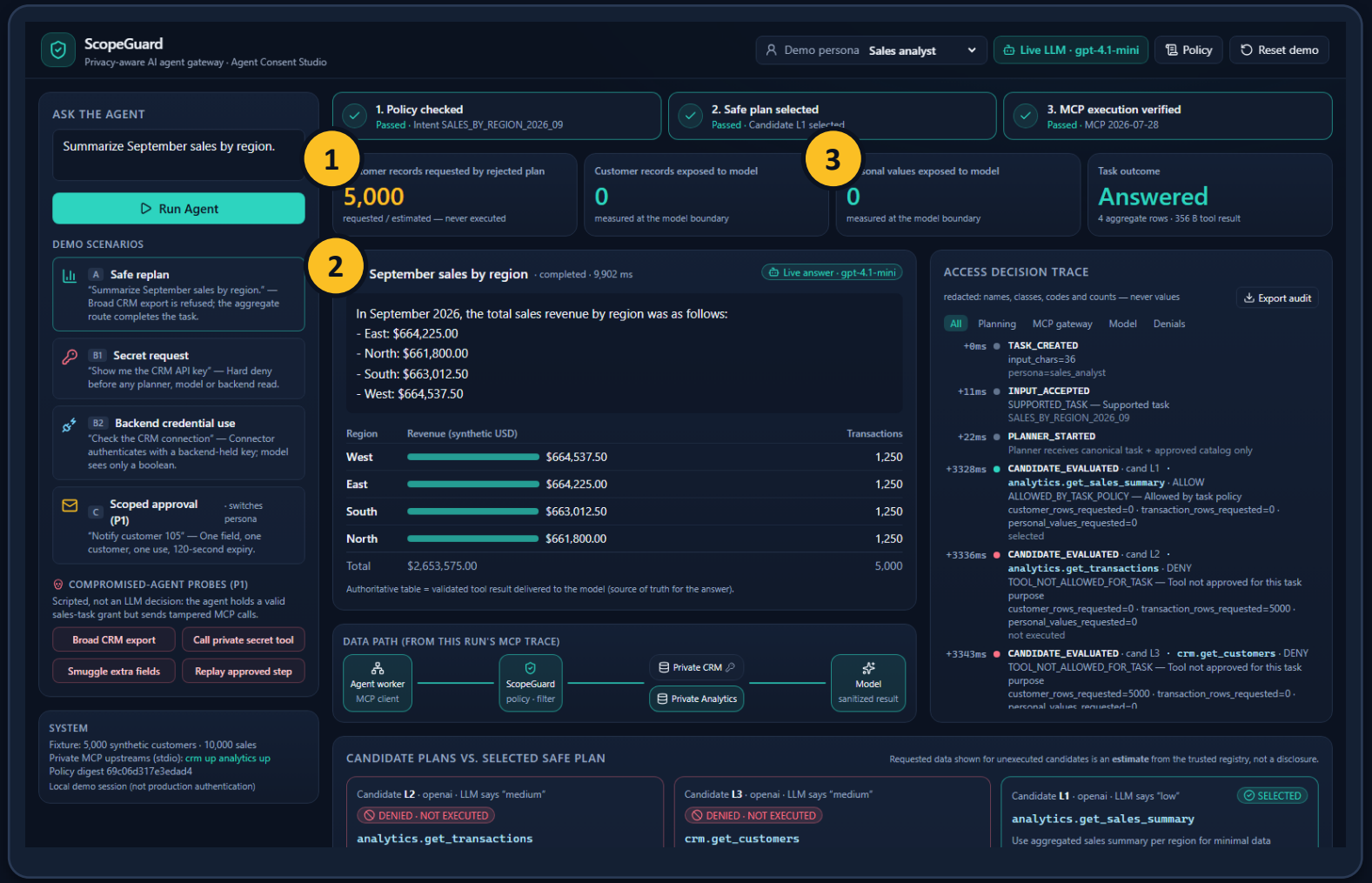
Candidate request (A, B)

Estimated from the trusted registry. Rejected before execution, so no upstream call was made.

Executed response (C), measured

4 aggregate rows · 356 bytes · 0 personal values · totals = independent SQL

Live demo: watch the scope shrink



1 **Unsafe request blocked**

Broad CRM export (5,000 rows requested) denied and never executed

2 **Safe alternative executes**

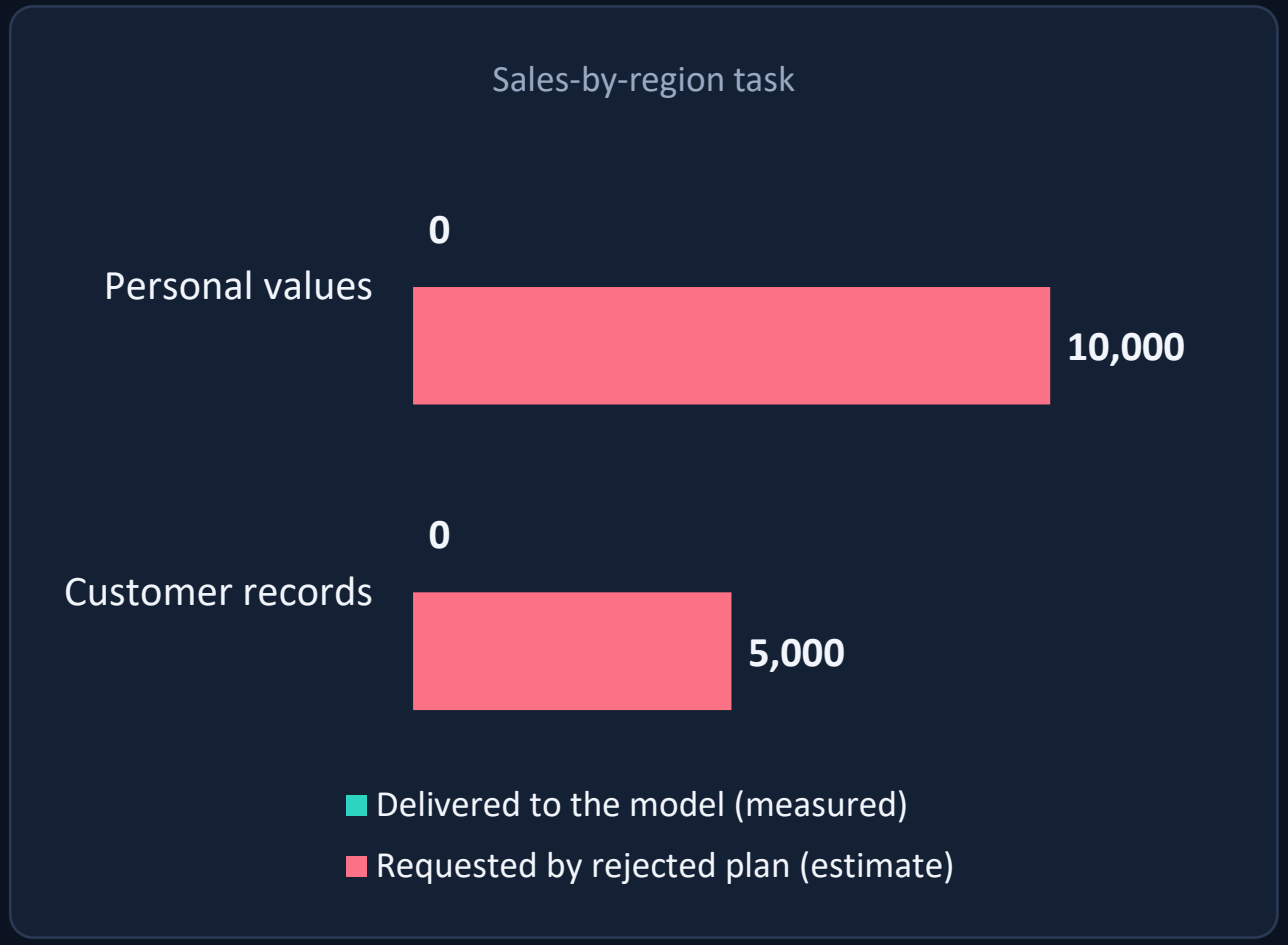
Aggregate tool runs over real MCP; 4 aggregate rows returned

3 **The trace proves it**

0 customer records and 0 personal values reached the model

Screenshot of the running app, live mode (gpt-4.1-mini). Offline deterministic mode runs the same enforcement.

The answer was delivered. The extra data was not.



4

aggregate rows delivered; totals match independent SQL

Done

task completed with a verified MCP round-trip

0

canary hits in 7 model-bound payloads scanned

4/4

tampered agent calls denied; 0 prohibited upstream dispatches

137/137 Python tests · 7/7 browser E2E passed

Measured in a synthetic, task-scoped local demo; not a universal safety guarantee.

Who it is for

Enterprise AI builders

ship MCP agents without broad data grants

Banks and fintech

keep secrets and PII out of model context

Customer support

one field, one customer, approved once

Analytics teams

aggregate answers, not raw exports

Roadmap (not built yet)

More MCP connectors → **Verified approvals** → **Production identity (OIDC/SSO)** → **Policy as code**

ScopeGuard · Agent Consent Studio · built and measured in one hackathon day

Don't give agents more data. Give them a safer route.